# Class 3 Tasks

1. -rwxr-xr-x --> -rw-r----- : the chmod number is 110 100 000 => 640
2. 644 --> +X for everyone while rw are the same => 644 -> -rw-r--r-- let's add +x for all -rwx-r-xr-x => 755
3. writable by owner and group but readable for everyone -rwxrwxr-- => 774
4. 110 100 000 => 640
5. A file only accessible to root , yes
6. 
```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ touch secret.log

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ chmod 000 secret.log

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ cat secret.log
cat: secret.log: Permission denied

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ sudo cat secret.log
[sudo] password for barok:

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ ls
12793.zip  flag4.txt  secret.log

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ ls -l
total 8
-rw-rw-r-- 1 barok barok 209 Mar 21 16:35 12793.zip
-rw-rw-r-- 1 barok barok  41 Mar 21 12:21 flag4.txt
---------- 1 barok barok   0 Mar 22 00:49 secret.log

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$
```
7. touch Audit.txt ; sudo chmod 750 Audit.txt;

```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ touch Audit.txt; sudo chmod 750 Audit.txt

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ ls -l
total 8
-rw-rw-r-- 1 barok barok 209 Mar 21 16:35 12793.zip
-rwxr-x--- 1 barok barok   0 Mar 22 00:50 Audit.txt
-rw-rw-r-- 1 barok barok  41 Mar 21 12:21 flag4.txt
---------- 1 barok barok   0 Mar 22 00:49 secret.log

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$
```

8. 741 --> -rwxr----x

9. 500 --> 101 --> r-x
10. -r-xrw-r-- => 101 110 100 --> 564

Task 2:

```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ exiftool ./challenge.jpg
xifTool Version Number         : 13.50
ile Name                       : challenge.jpg
irectory                       : .
ile Size                       : 76 kB
ile Modification Date/Time     : 2026:03:21 15:57:59-05:00
ile Access Date/Time           : 2026:03:21 16:05:07-05:00
ile Inode Change Date/Time     : 2026:03:21 15:57:59-05:00
ile Permissions                : -rw-rw-r--
ile Type                       : JPEG
ile Type Extension             : jpg
IME Type                       : image/jpeg
FIF Version                    : 1.01
esolution Unit                 : None
 Resolution                    : 1
 Resolution                    : 1
omment                         : FLAG{ALWAYS_Check_Metadata}
mage Width                     : 610
mage Height                    : 343
ncoding Process                : Baseline DCT, Huffman coding
its Per Sample                 : 8
olor Components                : 3
 Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
mage Size                      : 610×343
egapixels                      : 0.209
```

First flag in the Image Meta Data:
FLAG{ALWAYS_Check_Metadata}

```
E%BK
CG}{
; bw
lkc)S
zoi#
\M?a
ye$0
I;s#
}7N7@
yE|q
[{jv
+1<{
to}c
J+%R]
91DO
"kh\
SNKbZ
!rAt,F
)_+#
 F[d
h|M"
vv8iv
R?"#
;aWn
)J>_
E)Ih
8+B-lq:
8Xqo
+dF-
6sLZb
}@3_
.wj)-4
)a^v
MZN*
EV;2
NUi6
[M>s
FLAG{Strings_leak_information}
9:-0
*+&&',?PK
flag4.txtUT
Umt4QlIzdGhjSEJsYm1SbFpGOTZhWEI5Q2c9PQo=
flag4.txtUT

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ cat challenge.jpg | strings | grep "FLAG"
FLAG{ALWAYS_Check_Metadata}
FLAG{Strings_leak_information}

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$
```

the second flag was hidden inside the binary file in clear view and found using cat, strings and grep:

FLAG{Strings_leak_information}

```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ ls /usr/share/wordlists
dirb       dnsmap.txt     fern-wifi  legion      nmap.lst        sqlmap.txt  wifite.txt
dirbuster  fasttrack.txt  john.lst   metasploit  rockyou.txt.gz  wfuzz

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ stegseek --seed challenge.jpg --wordlist /usr/share/wordlists/rockyou.txt.gz
[!] error: the argument "--wordlist" is not compatible with the current command
stegseek: type "stegseek --help" for help.

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ gzip -d /usr/share/wordlists/rockyou.txt.gz

gzip: /usr/share/wordlists/rockyou.txt: Permission denied

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ cd /usr/share/wordlists/ ; sudo gzip -d ./rockyou.txt.gz;

┌──(barok㉿kali)-[/usr/share/wordlists]
└─$ cd -
~/Downloads/Telegram Desktop

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ stegseek challenge.jpg /usr/share/wordlists/rockyou.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "password123"
[i] Original filename: "flag5.txt".
[i] Extracting to "challenge.jpg.out".


┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$
```

```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ cat challenge.jpg.out
FLAG{steghide_protected}

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$
```

this is the third flag i found by trying to extract the passphrase from this encrypted file using brute force from rockyou.txt wordlist using cmd stegseek :

FLAG{steghide_protected}

```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ binwalk -eMr challenge.jpg

Scan Time:     2026-03-21 16:41:43
Target File:   /home/barok/Downloads/Telegram Desktop/challenge.jpg
MD5 Checksum:  2da1e6db0a36e3de001d42e13f69aacc
Signatures:    436

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
75667         0x12793         Zip archive data, at least v1.0 to extract, compressed size: 41, uncompressed size: 41,
 name: flag4.txt

WARNING: One or more files failed to extract: either no utility was found or it's unimplemented


Scan Time:     2026-03-21 16:41:44
Target File:   /home/barok/Downloads/Telegram Desktop/_challenge.jpg-2.extracted/flag4.txt
MD5 Checksum:  ff58b44f7e20695ab02140ba5bc2472b
Signatures:    436

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------



┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ ls
 challenge_1.txt              _challenge.jpg-2.extracted   linux-basic-for-hackers.pdf
 challenge_2.txt              _challenge.jpg.extracted     Screenshot_20260321-061337.jpg
 challenge.jpg                challenge.jpg.out            Screenshot_20260321-062120.jpg
 _challenge.jpg-0.extracted   flag5.txt
 _challenge.jpg-1.extracted  'Linux -adva v2.pdf'

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ cd _challenge.jpg-2.extracted

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg-2.extracted]
└─$ ls
flag4.txt

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg-2.extracted]
└─$ cat flag4.txt
Umt4QlIzdGhjSEJsYm1SbFpGOTZhWEI5Q2c9PQo=

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg-2.extracted]
└─$
```

```
┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ ls
 challenge_1.txt    _challenge.jpg.extracted   linux-basic-for-hackers.pdf
 challenge_2.txt    challenge.jpg.out          Screenshot_20260321-061337.jpg
 challenge.jpg     'Linux -adva v2.pdf'        Screenshot_20260321-062120.jpg

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop]
└─$ cd _challenge.jpg.extracted

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ ls
12793.zip  flag4.txt

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ cat flag4.txt | base64 -d
RkxBR3thcHBlbmRlZF96aXB9Cg==

┌──(barok㉿kali)-[~/Downloads/Telegram Desktop/_challenge.jpg.extracted]
└─$ cat flag4.txt | base64 -d | base64 -d
FLAG{appended_zip}
```

this flag was found embedded in the image file as a zip and found using the tool binwalk : FLAG{appended_zip}